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APPLIED ML & FINTECH INTELLIGENCE · CASE STUDY

AIOps Expense Tracker: NGN ML Categoriser & Forecaster

BI Scientist • AI/ML Engineer • Python Developer

Financial intelligence platform tailored to Nigerian bank statements (GTBank, Access, OPay, Kuda) with 0.9204 Macro-F1, receipt OCR, and ETS spend forecasting.

NIGERIAN BANKING NUANCE

Why Global Personal Finance Apps Fail in Nigeria

Standard personal finance apps rely on US/EU open-banking APIs and collapse all Nigerian transactions into generic 'Bank Transfer' buckets.

Cryptic Narration Strings

- Narrations like `POS PUR NNPC LEKKI` or `USSD/MTN DATA` fail standard classifiers.
- Abbreviations, merchant terminal codes, and bank fees are mislabeled.
- Users abandon apps that require 80% manual re-categorization.

Lumpy Outlay Anomaly Skew

- Annual rent or bulk generator fuel purchases distort standard z-scores.
- Global anomaly models trigger hundreds of false alarms.
- Need per-category Median Absolute Deviation (MAD) baselines.

Volatile Inflation Spikes

- High inflation causes routine monthly budget ceilings to break.
- Need seasonal forecasting with explicit prediction intervals.
- Must account for salary-week purchasing patterns.

MACHINE LEARNING ARCHITECTURE

TF-IDF N-Grams, Receipt OCR & Holt-Winters ETS

Hybrid machine learning architecture engineered specifically for Nigerian financial data.

Hybrid Categoriser

- Word + Char-WB TF-IDF (3–5 n-grams).
- Calibrated Logistic Regression.
- Cold-start keyword rule fallback.
- Online sample-weighted retraining.

Multi-Bank Ingestion

- Parsers for GTBank, Access, OPay, Kuda, Moniepoint.
- Regex / OCR parser for receipts & POS slips.
- 97.5% field exact-match rate.

MAD Anomaly & ETS

- Per-category MAD robust z-scores ($z \geq 2.5$).
- Damped Holt-Winters ETS 3-month forecast.
- Holdout WAPE = 20.42%.

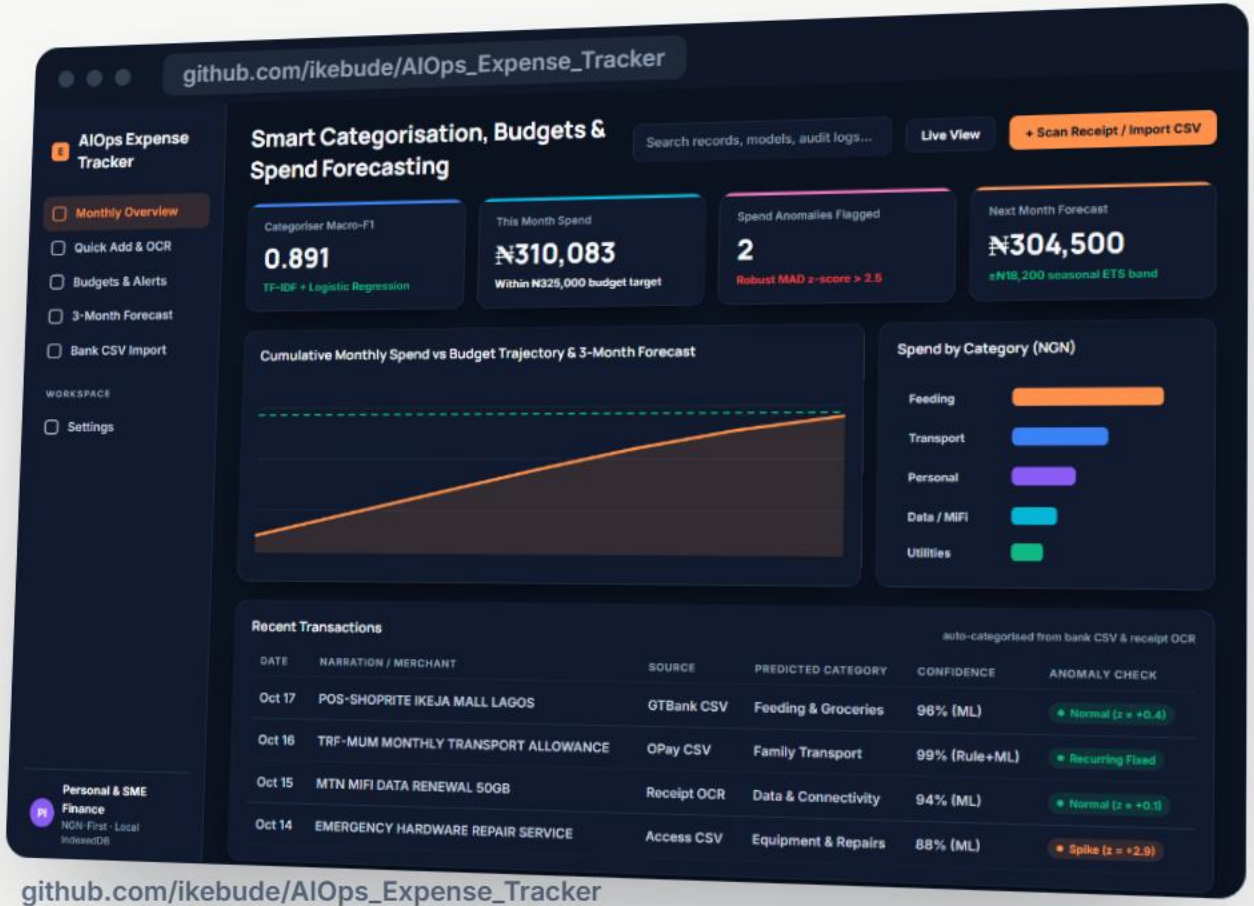
APPLICATION DASHBOARD

AIOps Expense Management App

React + Vite frontend powered by a FastAPI machine learning backend.

- **Native Nigerian Bank Import:** Drop in any GTBank, Access Bank, OPay, or Kuda statement CSV to auto-parse and auto-categorise.
- **Receipt Scanner:** Extract merchant, date, total, and VAT from supermarket receipts and POS charge slips.
- **Adaptive Learning:** When a user corrects a category, the model immediately performs online sample-weighted retraining.

Scikit-learn | FastAPI | React / Vite | Holt-Winters



github.com/ikebude/AIOps_Expense_Tracker

VERIFIED METRICS

Rigorous Evaluation on 600 Transactions

Verified results on 600 labelled Nigerian transactions across 10 categories (`reports/metrics.json`).

Macro-F1	Accuracy	Receipt OCR Match	Forecast WAPE
0.9204	92.67%	97.50%	20.42%
95% CI: 0.859–0.961	95% CI: 0.880–0.967	100% on Merchant, Date, Total	MAE ₦55,523 vs ₦74,208 baseline

Lift Over Heuristics

- Rule-only keyword baseline achieved 0.7801 Macro-F1.
- The hybrid ML model delivers a +14.03% lift in Macro-F1.
- Rule fallback rate on the test set is only 2.0%.

Robust Anomaly Detection

- 100% recall on out-of-band anomalies (e.g. ₦5,577 ATM surcharge vs ₦262 median).
- Category-level MAD prevents large legitimate expenses like rent from triggering false alerts.
- 9/9 Pytest tests passing in automated GitHub Actions CI.

OPEN-SOURCE & AFRICAN FINTECH

Personal finance intelligence built for Nigerian reality.

AIOps Expense Tracker combines lightweight NLP with local financial context to make automated budgeting work seamlessly in Africa.

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